

Cohort Definition Examples

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```
library(Capr)
```

This vignette provides a number of example cohorts built with Capr from the OHDSI *Phenotype Phebruary* event.

1 Type 2 diabetes mellitus

Persons with new type 2 diabetes mellitus at first diagnosis

<https://atlas-phenotype.ohdsi.org/#/cohortdefinition/88/definition>

```
cs0 <- cs(descendants(443238, 201820, 442793),
           descendants(exclude(195771, 201254, 435216, 761051, 4058243, 40484648)),
           name = "Type 2 diabetes mellitus (diabetes mellitus excluding T1DM and secondary)")

ch <- cohort(
  entry = entry(
    conditionOccurrence(cs0),
    observationWindow = continuousObservation(priorDays = 365)
  ),
  exit = exit(
    endStrategy = observationExit()
  )
)

chJson <- toCohortJson(ch)
```

Persons with new type 2 diabetes and no prior T1DM or secondary diabetes

<https://atlas-phenotype.ohdsi.org/#/cohortdefinition/89/export>

```
cs0 <- cs(descendants(443238, 201820, 442793),
           descendants(exclude(195771, 201254, 435216, 761051, 4058243, 40484648)),
           name = "Type 2 diabetes mellitus (diabetes mellitus excluding T1DM and secondary)")

cs1 <- cs(descendants(201254, 435216, 40484648),
```

```

      name = "Type 1 diabetes mellitus")

cs2 <- cs(descendants(195771),
      name = "Secondary diabetes mellitus")

ch <- cohort(
  entry = entry(
    conditionOccurrence(cs0),
    observationWindow = continuousObservation(priorDays = 365)
  ),
  attrition = attrition(
    't1d' = withAll(
      exactly(0, conditionOccurrence(cs1), duringInterval(eventStarts(-Inf, 0)))
    ),
    'secondaryDiabetes' = withAll(
      exactly(0, conditionOccurrence(cs2), duringInterval(eventStarts(-Inf, 0)))
    )
  ),
  exit = exit(
    endStrategy = observationExit()
  )
)

chJson <- toCohortJson(ch)

```

Persons with new type 2 diabetes mellitus at first dx rx or lab

<https://atlas-phenotype.ohdsi.org/#/cohortdefinition/90>

```

cs0 <- cs(descendants(443238, 201820, 442793),
      descendants(exclude(195771, 201254, 435216, 761051, 4058243, 40484648)),
      name = "Type 2 diabetes mellitus (diabetes mellitus excluding T1DM and secondary)")

cs1 <- cs(descendants(201254, 435216, 40484648),
      name = "Type 1 diabetes mellitus")

cs2 <- cs(descendants(195771),
      name = "Secondary diabetes mellitus")

cs3 <- cs(descendants(4184637, 37059902),
      name = "Hemoglobin A1c (HbA1c) measurements")

cs4 <- cs(descendants(21600744),
      name = "Drugs for diabetes except insulin")

ch <- cohort(
  entry = entry(
    conditionOccurrence(cs0),
    drugExposure(cs4),
    measurement(cs3, valueAsNumber(bt(6.5, 30)), measurementUnit(8554L)), # percent
    measurement(cs3, valueAsNumber(bt(48, 99)), measurementUnit(9579L)), # mmol/mol
    observationWindow = continuousObservation(priorDays = 365)
  ),
  attrition = attrition(

```

```

    'no T1D' = withAll(
      exactly(0, conditionOccurrence(cs1), duringInterval(eventStarts(-Inf, 0)))
    ),
    'no secondary diabetes' = withAll(
      exactly(0, conditionOccurrence(cs2), duringInterval(eventStarts(-Inf, 0)))
    )
  ),
  exit = exit(
    endStrategy = observationExit()
  )
)

chJson <- toCohortJson(ch)

```

2 Type 1 diabetes mellitus

Persons with new type 1 diabetes

<https://atlas-phenotype.ohdsi.org/#/cohortdefinition/92>

```

cs0 <- cs(descendants(195771),
  name = "Type 1 diabetes mellitus")

ch <- cohort(
  entry = entry(
    conditionOccurrence(cs0),
    observationWindow = continuousObservation(priorDays = 365)
  )
)

chJson <- toCohortJson(ch)

```

Persons with new type 1 diabetes and no prior T2DM or secondary diabetes

<https://atlas-phenotype.ohdsi.org/#/cohortdefinition/91>

```

cs0 <- cs(descendants(443238, 201820, 442793),
  descendants(exclude(195771, 201254, 435216, 761051, 4058243, 40484648)),
  name = "Type 2 diabetes mellitus (diabetes mellitus excluding T1DM and secondary)")

cs1 <- cs(descendants(201254, 435216, 40484648),
  name = "Type 1 diabetes mellitus")

cs2 <- cs(descendants(195771),
  name = "Secondary diabetes mellitus")

ch <- cohort(
  entry = entry(
    conditionOccurrence(cs1),
    observationWindow = continuousObservation(priorDays = 365)
  ),
  attrition = attrition(
    "no prior T2DM" = withAll(exactly(0, conditionOccurrence(cs0), duringInterval(eventStarts(-Inf, 0))),
    "no prior secondary T1DM" = withAll(exactly(0, conditionOccurrence(cs2), duringInterval(eventStarts(-Inf, 0)))
  )
)

```

```
)  
  
chJson <- toCohortJson(ch)
```

3 Atrial Fibrillation

Persons with atrial fibrillation per Wharton et al 2021

<https://atlas-phenotype.ohdsi.org/#!/cohortdefinition/93>

```
cs0 <- cs(descendants(313217),  
          name = "Atrial fibrillation")  
  
ch <- cohort(conditionOccurrence(cs0))  
  
chJson <- toCohortJson(ch)
```

Persons with atrial fibrillation per Subramanya et al 2021

<https://atlas-phenotype.ohdsi.org/#!/cohortdefinition/94>

```
afib <- cs(descendants(313217),  
          name = "Atrial fibrillation")  
  
ip <- cs(descendants(262, 9201),  
          name = "Inpatient or inpatient ER visit")  
  
op <- cs(descendants(9202, 9203),  
          name = "Outpatient or ER visit")  
  
ch <- cohort(  
  entry = entry(  
    conditionOccurrence(  
      conceptSet = afib,  
      nestedWithAll(  
        # an inpatient afib  
        atLeast(  
          x = 1,  
          aperture = duringInterval(eventStarts(-Inf, 0), eventEnds(0, Inf)),  
          query = visit(conceptSet = ip)  
        )  
      )  
    ),  
    conditionOccurrence(  
      conceptSet = afib,  
      nestedWithAll(  
        # 2 afib in outpatient  
        atLeast(  
          x = 1,  
          aperture = duringInterval(eventStarts(-Inf, 0), eventEnds(0, Inf)),  
          query = visit(  
            conceptSet = op,  
            nestedWithAll(  
              atLeast(  

```


- **Occ cohort** -> primaryCriteriaLimit = "All", expressionLimit = "All", era = era(eraDays = 99999L)

The two patterns below show different ways to get one row per person:

- **Era version:** take the first qualifying event and follow until end of continuous observation.
- **Occ version:** take all qualifying events and stitch them into one long era from first to last event.

5.1 Example 1: Nested criteria with prior diagnosis requirement

5.1.1 1a. Era version

```
# Placeholder concept IDs for disease of interest
disease_concept_set <- c(201826L)

# Index event: first diagnosis with a prior diagnosis at least 365 days earlier
diagnosisEraCohort <- cohort(
  entry = entry(
    conditionOccurrence(
      conceptSet = cs(disease_concept_set, name = "Disease"),
      nestedWithAll(
        atLeast(
          1,
          conditionOccurrence(conceptSet = cs(disease_concept_set, name = "Disease")),
          aperture = duringInterval(eventStarts(-Inf, -365))
        )
      )
    ),
    primaryCriteriaLimit = "First"
  ),
  attrition = attrition(
    expressionLimit = "First"
  ),
  exit = exit(
    endStrategy = observationExit()
  )
)

diagnosisEraJson <- toCohortJson(diagnosisEraCohort)
```

5.1.2 1b. Occ version (same cohort logic, different type)

```
# Entry at each diagnosis with a prior diagnosis at least 365 days earlier
diagnosisOccCohort <- cohort(
  entry = entry(
    conditionOccurrence(
      conceptSet = cs(disease_concept_set, name = "Disease"),
      nestedWithAll(
        atLeast(
          1,
          conditionOccurrence(conceptSet = cs(disease_concept_set, name = "Disease")),
          aperture = duringInterval(eventStarts(-Inf, -365))
        )
      )
    ),
  ),
```

```
    primaryCriteriaLimit = "All"  
  ),  
  attrition = attrition(  
    expressionLimit = "All"  
  ),  
  exit = exit(  
    endStrategy = fixedExit(offsetDays = 0L)  
  ),  
  era = era(eraDays = 99999L)  
)  
  
diagnosisOccJson <- toCohortJson(diagnosisOccCohort)
```